ExaThon India 2026 · Track 01: Money, Risk & Regulation

“...interest was charged from the date of the cheque whereas the cheque was handed over to the customer several days later.”

Reserve Bank of India, on what it found at lenders · April 2024



Sahi Byaaj

Finds every rupee of interest charged before the money arrived.

Team Hacksmiths
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All data in this deck is synthetic



She has no idea.

Meena paid 8 days of interest on money she didn’t have yet

8 days

3 Mar
Cheque dated. Interest starts.

11 Mar
Cheque actually handed over.

₹2,301.37

overcharged on one payment of ₹12,00,000 · synthetic borrower

Why nobody catches it



ONE CHEQUE, THREE RECORDS

Date on the cheque	**3 Mar**
Loan system	**3 Mar**
Builder’s handover receipt	**11 Mar**

The one that’s true decides the interest. Usually it sits outside the loan system.

The dates disagree

Each record says something different, and real books have millions of them.

The system checks itself

The loan system that calculates interest is the one being audited. Its reports repeat its own mistakes.

The amounts are small

A few days on one payment. Borrowers rarely notice, so the lender has to find it.

Six practices RBI told lenders to stop



	What lenders were doing	Named	We check
1	Charging interest from the loan approval date, not the day money was paid	Apr 2024	Yes
2	Charging from the cheque date when the cheque was handed over later	Apr 2024	Yes
3	Charging a full month when money moved mid-month	Apr 2024	Yes
4	Charging on the full amount after advance EMIs were collected	Apr 2024	Yes
5	Working out the EMI on the approved amount, not the amount paid out	Jul 2026	Yes
6	Changing the repayment schedule after a part-payment without telling the borrower	Jul 2026	Next

RBI told lenders to **refund the extra interest**. In July 2026 it wrote these into its supervisory directions for banks, NBFCs and housing finance companies.

Sahi Byaaj checks every loan in three steps

1

Find the real date

When did the money actually arrive? Checked against every record.

2

Recalculate

Work out the fair interest, and the exact rule the bank’s system used.

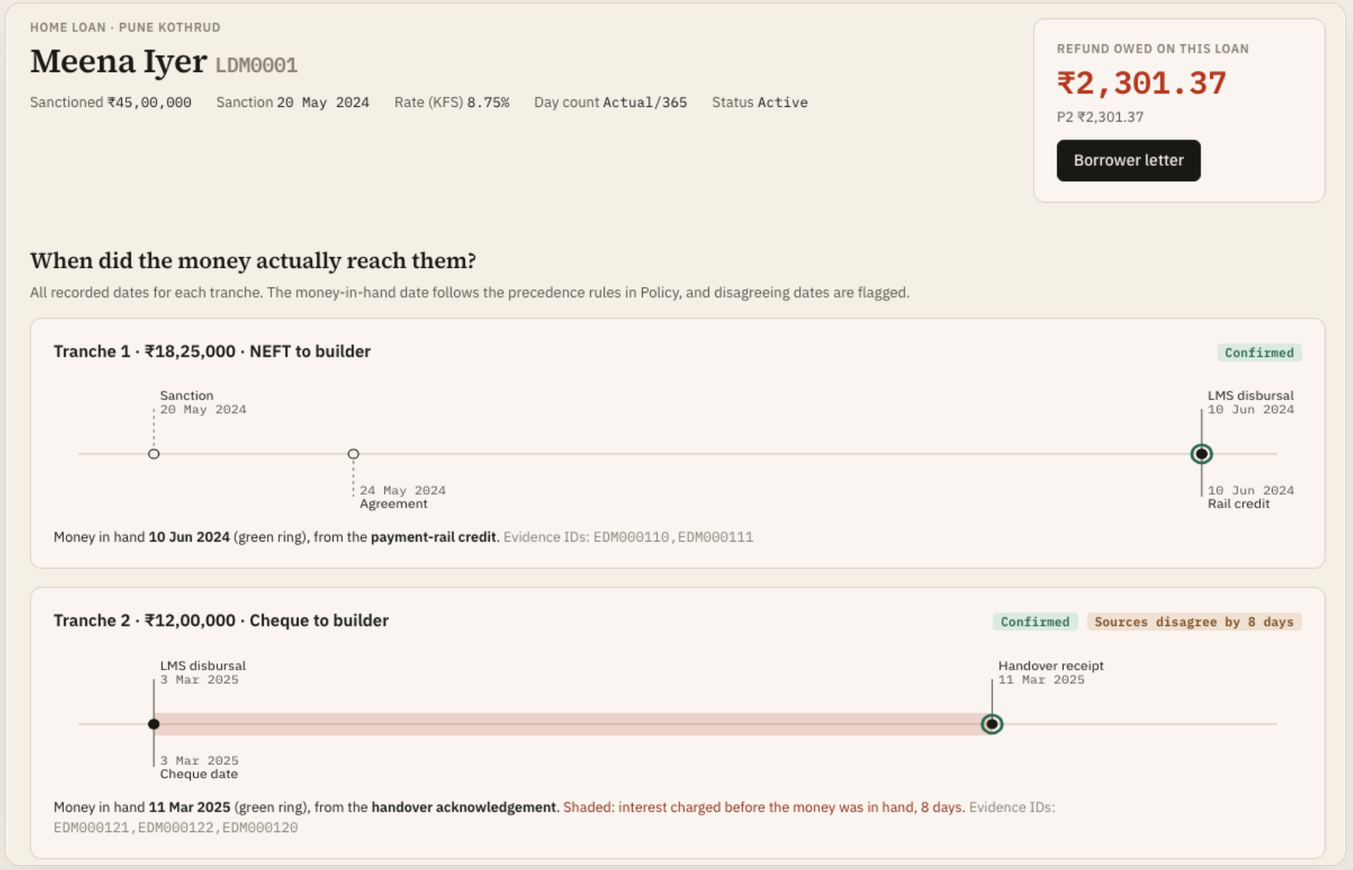
3

Refund with proof

The exact amount per borrower, with the evidence behind every rupee.

Whatever it can’t decide is **held out of the total**, and a query to the branch is written for it.

Every loan, explained on one screen



Every date, side by side

Disagreements are flagged.

The days overcharged

Shaded, with the matching rule.

The refund letter

English, Hindi or Tamil, every figure checked.

How it works out the rule the bank used

MEENA · MARCH 2025 · EVERY RULE RECALCULATED

What the bank's system charged	₹21,904.97	
From the day the money arrived (fair)	₹19,603.60	fair
From the sanction or agreement date	₹19,603.60	no match
From the cheque date	₹21,904.97	✓ exact
A full month for mid-month money	₹22,480.31	no match
On the full sanctioned ₹45,00,000	₹33,441.78	no match

Every rule that applies to this loan gives one of these four answers. Only one hits the bank's figure.

OVERCHARGED THIS MONTH

₹2,301.37

₹21,904.97 charged – ₹19,603.60 fair

```
₹12,00,000 × 8.75% × 8 days ÷ 365 = ₹2,301.37
```

The bank's figure is reproduced **exactly, to the paisa,** by one rule: **from the cheque date** P2. Nobody had to tell the tool; the posting itself is the evidence.

4 start dates × **2** loan amounts × **2** advance-EMI rules × **2** month roundings + **1** allowed fee = **33** checks

Every month is recalculated all 33 ways. The one that matches the bank's own figure **to the paisa** shows exactly what its system did. Nothing is assumed.

What happens to every loan, step by step

1

Read the records

Loan terms from the KFS, the cheque, handover and bank-credit dates, repayments, and **what the bank charged**.

2

Find the real date

The record that proves when the money arrived decides. **No proof? The loan goes on hold** and the branch is asked.

3

Recalculate

The fair interest from that date, and every wrong way the bank could have charged: **33 rules**, side by side.

4

Match to the paisa

The rule that gives **exactly the bank's figure** is what its system did. Excess is always charged minus fair.

MEENA, MARCH 2025, THROUGH THE SAME FOUR STEPS

Cheque **3 Mar**, handover **11 Mar**, bank charged **₹21,904.97**

Money in hand on **11 Mar**, from the handover receipt

Fair is **₹19,603.60**; the rules give four different answers

Only "from the cheque date" matches: refund **₹2,301.37**

Every month ends in one of five answers · only refunds go into the total

- **Fair** — Charged correctly. Nothing owed.
- **Refund** — Wrong rule named, exact amount owed.
- **Permitted** — A fee the KFS disclosed. Allowed.
- **On hold** — Date not proven. Branch asked.
- **Unexplained** — No rule fits. A person checks.

It also knows when not to refund

Raghav

₹0 refund

His contract disclosed this fee up front. It’s allowed, so no refund, and the reason is shown.

If the auditor decides the fee isn’t fair, one switch re-runs the book:

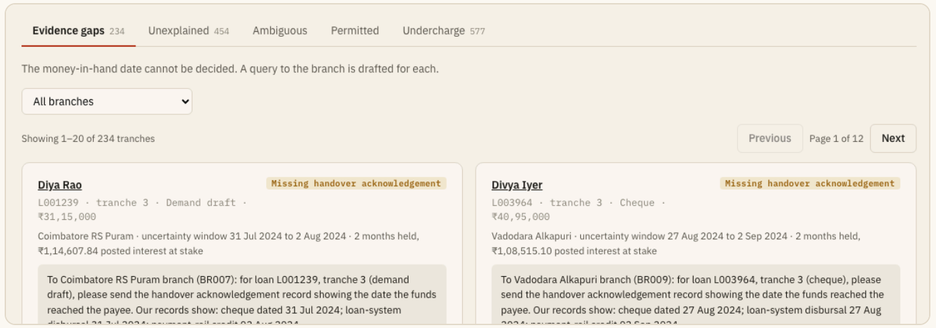
+₹30,68,322.19

for 250 more borrowers, in 16.5 s

Asha

On hold

Her records don’t settle the date. It writes to the branch instead of guessing.



DRAFTED FOR ASHA'S BRANCH, WORD FOR WORD

"To Pune Hadapsar branch (BR002): for loan LDM0003, tranche 1 (cheque), please send the **handover acknowledgement** record showing the date the funds reached the payee. Our records show: cheque dated 02 Jun 2025; loan-system disbursal 02 Jun 2025; payment-rail credit 06 Jun 2025."

One run over a 10,000-loan test book

₹8.83 crore

owed to 3,063 borrowers

exactly ₹8,83,23,995.69 · synthetic data

And everything it couldn’t decide is shown, not guessed:

Not included in the refund total

These need missing evidence or a reviewer's decision before any refund is made.

EVIDENCE GAPS

234 tranches

₹53,99,158 posted interest at stake. Branch queries drafted.

UNEXPLAINED POSTINGS

454 periods

81 loans where no known calculation method reproduces the posting. Needs manual review.

PERMITTED, NOT REFUNDED

₹30,68,322

250 loans charging a commitment fee the KFS disclosed. Policy allows it.

AMBIGUOUS, INCLUDED

₹47,97,684

The overcharge is certain, but more than one practice explains it. Counted in the total by policy.

Did the bank’s fix reach old loans too?

Was it fixed for everyone, or only for new loans?

Shows whether a loan-system fix reached loans already on the book, or only loans sanctioned after it.

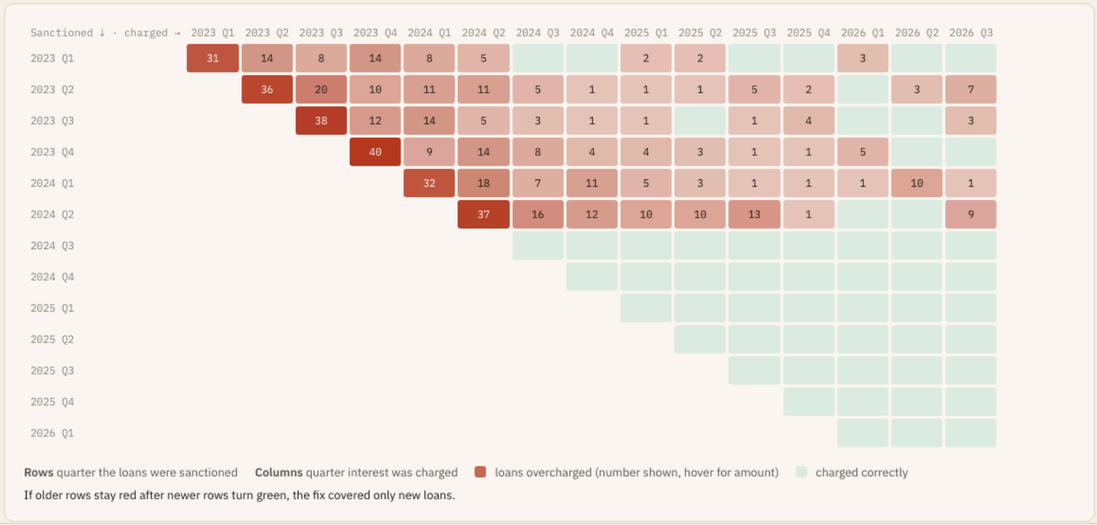
Nagpur Civil Lines (BR003)

256 loans sanctioned Jan 2023 to Jun 2024 charged interest from the money-in-hand date, rounded out to whole months, last in **Sep 2026**. Excess ₹38,34,987.13.

12 loans sanctioned Feb 2023 to Jun 2024 charged interest from the cheque date, rounded out to whole months, last in **Aug 2026**. Excess ₹3,61,580.49.

2 loans sanctioned May 2023 to Mar 2024 charged interest from the sanction date, last in **Apr 2024**. Excess ₹67,090.32.

1 loan sanctioned in Apr 2023 charged interest from the money-in-hand date, on the full sanctioned amount, last in **Sep 2023**. Excess ₹26,388.56.



Rows quarter the loans were sanctioned **Columns** quarter interest was charged loans overcharged (number shown, hover for amount) charged correctly

If older rows stay red after newer rows turn green, the fix covered only new loans.

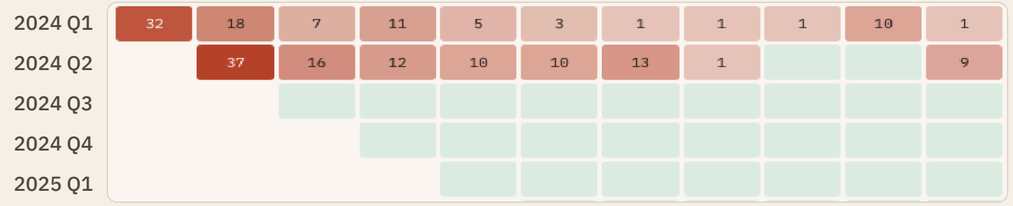
Rows: loans approved in a quarter.
Columns: the quarter interest was charged.
Red overcharged · **green** correct.

This branch fixed its system in July 2024. New loans turned green. **Older loans stayed red into 2026.**

For the head of operations

Re-run after every system change. A fix that skipped old loans shows up straight away.

ZOOM: AROUND THE JULY 2024 FIX



Approved before the fix: still charged in 2026. Approved after it: green from day one.

All the maths runs inside Exasol



Loan data
CSV or ZIP

→

Import
into Exasol

→

9 SQL steps
every rupee computed

→

Results
refund ledger

→

App, letters, AI
read only

Periods, not days

A balance changes a few dozen times per loan, so Exasol works on time periods: 4.2M for 100k loans, not a row per day.

All 33 checks at once

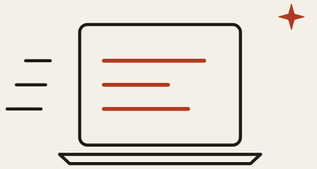
Each rule is stored only as its difference from the fair calculation, so all 33 run side by side in one pass.

Exact to the paisa

Whole-number maths in paise, rounded once per month. No floating point anywhere.

The AI assistant asks questions through Exasol’s MCP server with a read-only login. Exasol itself blocks it from the hidden test answers.

Half a million loans in under 8 minutes, on a laptop





11.6M

loan-months recalculated

2 CPUs

Exasol Personal, single node, on a MacBook Air

0 paise

off the checker on a 3,003-loan sample

Measured, not projected: 10k and 100k on an Apple M4 MacBook (16 GB), 500k on a MacBook Air M4 (24 GB); Exasol Personal VM with 2 vCPU on both. Source: docs/STATUS.md.

We tried hard to prove it wrong

0 paise

off an independent checker

Written separately in Python. Every month of the 10k book; samples at 100k and 500k.

17.07%

RBI’s own example, reproduced

₹20,000 over 24 months: EMI ₹970, interest ₹3,274 and APR match RBI’s published figures.

0

false alarms

Allowed fees and correctly charged months never reach the refund list.

98.37%

found, with messy data

100% on clean data, 96.82% with 3× the errors. No fair month refunded.

Same total

fresh install, second laptop

Following only the README, the refund total matched to the paisa.

26

automated tests passing

Plus automatic checks on every code push.

Who it helps, and how a lender uses it

Auditors

Know every overcharged borrower, with evidence, before the regulator does.

Borrowers

Get the exact refund, with a letter in English, Hindi or Tamil.

Lenders

Re-check the whole book after every system change.

Export loan data using our template → Upload a ZIP; it’s checked before loading → Whole book audited inside Exasol → Inspection pack, refunds, letters

Built for banks, NBFCs and housing finance companies under RBI’s directions. Home loans paid in stages carry most of the risk; the engine works for any loan.

Sahi Byaaj

Correct interest, for every borrower.

github.com/aryankalra404/Exasol

Open source (MIT) · runs on Exasol Personal · one command: make demo

Team Hacksmiths · Aashman Biyani · Aryan Kalra

> **Honest limits.** Synthetic data only. Not legal advice: disputed interpretations are policy settings. Refunds are slightly understated on purpose (no re-amortisation). Unproven dates are held, never guessed.

Code, run guide and docs
github.com/aryankalra404/Exasol